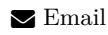


SHUWEI ZHOU

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Email



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RESEARCH PROFILE

Ph.D. candidate in computational mechanics at RWTH Aachen University, working at the interface of **fatigue and fracture mechanics**, **crystal plasticity**, and **physics-informed machine learning**. My research develops physics-structured and interpretable computational methods for metallic materials, with emphasis on microstructure-sensitive fatigue assessment, crack growth modeling, elastic/fracture field reconstruction, and damage-tolerant design.

Research Keywords: Fatigue and fracture mechanics; crystal plasticity; physics-informed machine learning; neural operators; metallic materials.

EDUCATION

RWTH Aachen University Oct. 2023 - Feb. 2027 (expected)
Ph.D. Candidate, Institute of Metal Forming Aachen, Germany
Supervisor: Prof. Dr.-Ing. Sebastian Münstermann
Topic area: physics-informed machine learning, Multiscale modeling, fatigue and fracture mechanics

Southwest Jiaotong University Sep. 2020 - Jun. 2023
M.Eng. in Vehicle Engineering Chengdu, China
State Key Laboratory of Rail Transit Vehicle System
Supervisors: Prof. Bing Yang and Prof. Shoune Xiao

Southwest Jiaotong University Sep. 2016 - Jun. 2020
B.Eng. in Engineering Mechanics Chengdu, China
School of Mechanics and Aerospace Engineering
Supervisor: Prof. Qianhua Kan

RESEARCH EXPERIENCE

Institute of Metal Forming, RWTH Aachen University Oct. 2023 - Present
Ph.D. Researcher, RWTH Aachen University Aachen, Germany

- Developed physics-informed neural networks for fatigue life prediction of metallic materials by embedding experiment-inspired fatigue constraints.
- Established CPFEM-fatigue indicator workflows for ferrite-pearlite steel, linking local cyclic deformation, fatigue crack initiation, and microstructurally short crack growth.
- Developed Kolosov-Muskhelishvili informed neural networks for elasticity/fracture field reconstruction, stress intensity factor identification, and crack propagation modeling.
- Performed large-scale Abaqus, DAMASK, and PyTorch computations on RWTH CLAIX using Linux and Slurm workflows.

State Key Laboratory of Rail Transit Vehicle System Sep. 2020 - Jun. 2023
Master's Research, Southwest Jiaotong University Chengdu, China

- Investigated fatigue crack growth, mixed mode fracture, and crack closure behavior of railway steels and aluminum alloys.
- Conducted fatigue and fracture tests, mechanical testing, and microstructural characterization of 6005A-T6 and 7020-T6 aluminum alloys.
- Studied both base metals and welded joints, and developed machine learning, bearing equivalent test, and finite element methods for fatigue life prediction and structural assessment.

TECHNICAL SKILLS

- Experimental mechanics:** tensile testing, fatigue testing, fracture mechanics testing, and DIC.
- Materials characterization:** SEM/EDS, EBSD, CLSM, OM, and hardness testing.
- Computational modeling:** finite element analysis, crystal plasticity, fracture mechanics, and multiscale modeling.
- Software and HPC:** Abaqus, Ansys, Nastran, HyperMesh, DAMASK, MTEX, and Linux.
- Scientific ML and coding:** Python, PyTorch, MATLAB, Fortran, Linux, Git, LaTeX, and Slurm-based HPC workflows.

PUBLICATIONS

19 peer-reviewed publications, including 7 first-author journal articles and 12 additional collaborative articles.
ORCID: 0000-0002-7801-909X **Scopus ID:** 57226158345 **Google Scholar:** 285 citations; h-index: 10

First-author publications

Shuwei Zhou, Christian Häffner, Sophie Stebner, Niklas Fehlemann, Zhichao Wei, Sebastian Münstermann. (2026). “A Variational Kolosov-Muskhelishvili Network for Elasticity and Fracture”, *International Journal of Mechanical Sciences*, 111860. (IF: 11.4); DOI: [10.1016/j.ijmecsci.2026.111860](https://doi.org/10.1016/j.ijmecsci.2026.111860)

Shuwei Zhou, Christian Häffner, Shuancheng Wang, Sophie Stebner, Zhen Liao, Bing Yang, Zhichao Wei, Sebastian Münstermann. (2026). “Transfer-learned Kolosov-Muskhelishvili Informed Neural Networks for Fracture Mechanics”, *Theoretical and Applied Fracture Mechanics*, 144, 105582. (IF: 6.0); DOI: [10.1016/j.tafmec.2026.105582](https://doi.org/10.1016/j.tafmec.2026.105582)

Shuwei Zhou, Mian Huang, Christian Häffner, Sophie Stebner, Min Cai, Zhichao Wei, Bing Yang, Sebastian Münstermann. (2026). “Microstructure-sensitive crystal plasticity and fatigue indicator modeling for LZ50 steel”, *International Journal of Fatigue*, 203, 109302. (IF: 7.0); DOI: [10.1016/j.ijfatigue.2025.109302](https://doi.org/10.1016/j.ijfatigue.2025.109302)

Shuwei Zhou, Manuel Henrich, Zhichao Wei, Feng Feng, Bing Yang, Sebastian Münstermann. (2025). “A general physics-informed neural network framework for fatigue life prediction of metallic materials”, *Engineering Fracture Mechanics*, 322, 111136. (IF: 6.2); DOI: [10.1016/j.engfracmech.2025.111136](https://doi.org/10.1016/j.engfracmech.2025.111136)

Shuwei Zhou, Bing Yang, Shoune Xiao, Guangwu Yang, Tao Zhu. (2024). “Interpretable Machine Learning Method for Modelling Fatigue Short Crack Growth Behaviour”, *Metals and Materials International*, 30(7), 1944–1964. (IF: 4.5); DOI: [10.1007/s12540-024-01628-6](https://doi.org/10.1007/s12540-024-01628-6)

Shuwei Zhou, Bing Yang, Shoune Xiao, Guangwu Yang, Tao Zhu. (2023). “Crack Growth Rate Model Derived from Domain Knowledge-Guided Symbolic Regression”, *Chinese Journal of Mechanical Engineering*, 36(1). (IF: 5.1); DOI: [10.1186/s10033-023-00876-8](https://doi.org/10.1186/s10033-023-00876-8)

Shuwei Zhou, Bing Yang, Chao Wang, Shuancheng Wang, Shoune Xiao, Guangwu Yang, Tao Zhu. (2023). “Estimating the fatigue crack growth rate of 6005A-T6 aluminum alloys with different stress ratios using machine learning methods”, *Chinese Journal of Nonferrous Metals*, 33(8), 2416–2427. ; DOI: [10.11817/j.ysxb.1004.0609.2022-43508](https://doi.org/10.11817/j.ysxb.1004.0609.2022-43508)

Selected collaborative publications

Zhichao Wei, Yijia Mo, **Shuwei Zhou**, Steffen Gerke, Sebastian Münstermann, Michael Brüning. (2026). “Deep learning characterization of stress-state-dependent anisotropic ductile damage”, *International Journal of Mechanical Sciences*, 315, 111424. (IF: 11.4); DOI: [10.1016/j.ijmecsci.2026.111424](https://doi.org/10.1016/j.ijmecsci.2026.111424)

Mian Huang, Bing Yang, Yining Wang, Cheng Li, **Shuwei Zhou**, Dongdong Chen, Shoune Xiao. (2026). “Quasi-static compression energy absorption characteristics of bamboo-inspired heterogeneous hybrid structure with CFRP/SLM-fabricated AlSi10Mg lattice”, *Thin-Walled Structures*, 115160. (IF: 8.3); DOI: [10.1016/j.tws.2026.115160](https://doi.org/10.1016/j.tws.2026.115160)

Feng Feng, Tao Zhu, Bing Yang, Zhe Zhang, **Shuwei Zhou**, Shoune Xiao. (2025). “Probabilistic fatigue life prediction in additive manufacturing materials with a physics-informed neural network framework”, *Expert Systems with Applications*, 275, 127098. (IF: 9.4); DOI: [10.1016/j.eswa.2025.127098](https://doi.org/10.1016/j.eswa.2025.127098)

Weiyang Wen, Huan Qi, Zhen Liao, Lanxin Jiang, Mian Huang, Bing Yang, Zhiwei Wu, **Shuwei Zhou**. (2025). “In-situ study of crack propagation mechanisms in wire arc additive manufactured Al-Mg alloys: Multiple strategies involving microdefects and microstructures”, *Materials Science and Engineering: A*, 924, 147802. (IF: 7.9); DOI: [10.1016/j.msea.2025.147802](https://doi.org/10.1016/j.msea.2025.147802)

Shuancheng Wang, **Shuwei Zhou**, Bing Yang, Shoune Xiao, Guangwu Yang, Tao Zhu. (2024). “Effective stress intensity factor range for fatigue cracks propagating in mixed mode I-II loading”, *Engineering Fracture Mechanics*, 312, 110641. (IF: 6.2); DOI: [10.1016/j.engfracmech.2024.110641](https://doi.org/10.1016/j.engfracmech.2024.110641)

Shuancheng Wang, Bing Yang, **Shuwei Zhou**, Yuanzhi Wang, Shoune Xiao. (2024). “Effect of stress ratio and overload on mixed-mode crack propagation behaviour of EA4T steel”, *Engineering Fracture Mechanics*, 306, 110210. (IF: 6.2); DOI: [10.1016/j.engfracmech.2024.110210](https://doi.org/10.1016/j.engfracmech.2024.110210)

Shuancheng Wang, Bing Yang, **Shuwei Zhou**, Jian Li, Shoune Xiao. (2024). “Closure Effect of I+II Mixed-mode Crack for EA4T Axle Steel”, *Chinese Journal of Mechanical Engineering*, 37(1). (IF: 5.1); DOI: [10.1186/s10033-024-01061-1](https://doi.org/10.1186/s10033-024-01061-1)

Feng Feng, Tao Zhu, Bing Yang, **Shuwei Zhou**, Shoune Xiao. (2024). “A physics-informed neural network approach for predicting fatigue life of SLM 316L stainless steel based on defect features”, *International Journal of Fatigue*, 188, 108486. (IF: 7.0); DOI: [10.1016/j.ijfatigue.2024.108486](https://doi.org/10.1016/j.ijfatigue.2024.108486)

Shuancheng Wang, Bing Yang, Jian Li, **Shuwei Zhou**, Shoune Xiao. (2023). “Mixed mode crack growth behaviour considering plasticity-induced and roughness-induced closure”, *Engineering Fracture Mechanics*, 289, 109430. (IF: 6.2); DOI: [10.1016/j.engfracmech.2023.109430](https://doi.org/10.1016/j.engfracmech.2023.109430)

Bing Yang, Zhanjiang Wei, Zhen Liao, **Shuwei Zhou**, Shoune Xiao, Tao Zhu, Guangwu Yang, Mingmeng Wang. (2021). “Optimisation Method for Determination of Crack Tip Position Based on Gauss-Newton Iterative Technique”, *Chinese Journal of Mechanical Engineering*, 34(1). (IF: 5.1); DOI: [10.1186/s10033-021-00585-0](https://doi.org/10.1186/s10033-021-00585-0)

ADDITIONAL INFORMATION

Languages

English: professional working proficiency; German: basic knowledge; Chinese: native.

Peer review

Regular reviewer for 17 peer-reviewed journals in fatigue and fracture mechanics, computational mechanics, materials science, and artificial intelligence:

International Journal of Fatigue

Engineering Fracture Mechanics

Theoretical and Applied Fracture Mechanics

International Journal for Numerical Methods in Engineering

Reliability Engineering and System Safety

Expert Systems with Applications

Engineering Structures

Engineering Applications of Artificial Intelligence

Journal of Materials Research and Technology

Structural Health Monitoring

International Journal of Pressure Vessels and Piping

International Journal of Cast Metals Research

Journal of Central South University

Digital Engineering

Computers, Materials & Continua

Journal of Engineering Thermophysics

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